Three Generations of Healthcare IT: From the Digital Record to the Computable Care Process

Alexander Apartsin[1], Yehudit Aperstein[2]

[1]School of Computer Science, Holon Institute of Technology (HIT), Holon, Israel · [2]Intelligent Systems, Afeka Academic College of Engineering, Tel Aviv, Israel

Position paper

Abstract

Abstract Healthcare information technology can be analyzed more durably through the computational objects it supports than through individual technologies, which evolve rapidly. Read this way, the first generation made the clinical *record* computable, and the second made the clinical *fact* computable through controlled terminologies, structured systems, interoperability standards, and clinical natural-language processing. We propose that an emerging third layer makes patient-specific clinical *intent* increasingly computable from clinical communication: what should happen next, when, under which conditions, and by whom, information that today lives in free-form notes, discharge instructions, referrals, handovers, and messages. The claim is deliberately narrow. Existing workflow standards and guideline representations can encode care processes once these have been explicitly structured; the unresolved gap is upstream, because much patient-specific intent is expressed only in natural communication and therefore never enters a computable process layer. We define the Actionable Clinical Record (ACR) as the atomic object of that layer: a source-grounded representation of an intended clinical action together with the semantic attributes required to interpret, execute, monitor, or audit it. We give explicit criteria for a computational layer, distinguish prescribed, observed, and intended process, and formulate reliability as an explicit system property. We ground the argument in a controlled feasibility instance, follow-up extraction, in which a hybrid neural-symbolic pipeline reaches action–time Pair F1 of 0.997 while general-purpose LLMs reach about 0.53, and we set out testable predictions and a four-part Computable Care Process research agenda.

Keywords: healthcare IT; electronic health records; unit of computability; actionable clinical record; clinical intent; large language models; neural-symbolic AI; temporal reasoning; care-process automation; FHIR workflow

1. Introduction

Healthcare information technology can be analyzed more durably through the computational objects it supports than through individual technologies, which evolve rapidly. Individual technologies turn over quickly, whereas the computational objects they enable are more persistent. Organized this way, the field's history has a compact form: the first generation made clinical information available to computers; the second made clinical state interpretable by computers; the third, we propose, makes clinical intent operational by computers. Equivalently, the first made the *record* computable, the second made the *fact* computable, and the third makes patient-specific *clinical intent* increasingly computable from communication (Figure 1). Generational language has been used before for electronic medical records and for the twenty-five-year evolution of health IT;[1,2] the organizing principle we propose, the unit of computability paired with its enabling computational stack, is a synthesis rather than a restatement, because each transition unlocked a distinct class of computation, required a distinct stack, and served a distinct set of clinical needs. We use *generation* to denote the historical emergence of a new computational layer; the resulting layers coexist and accumulate rather than replace one another. An important asymmetry follows: generations one and two are retrospective abstractions over a well-documented history, whereas generation three is a prospective hypothesis, and we treat it as such throughout.

A persistent limitation of the first two layers is visible in routine clinical communication. A physician writes "repeat the CBC in two weeks" or "refer to cardiology if symptoms persist," and that instruction, though clinically decisive, is captured only as narrative text. The record stores it (generation one) and may code the associated diagnosis or medication (generation two), but the action, its timing, its condition, and its owner remain outside the computable substrate. The consequence is measurable: missed test-result follow-up ranges from 6.8% to 62%,[3] roughly half of recommended imaging follow-ups are never performed,[4] and only about a third of referrals reach a completed specialist visit.[5]

1.1 Scope and relation to existing process representations

Healthcare IT could represent process before large language models, whenever a person or system had already structured that process. FHIR represents patient-specific intent through *CarePlan* and *ServiceRequest* and generic plans through *PlanDefinition*;[15] computer-interpretable guidelines have encoded executable protocols for decades;[35,36] and process-mining pipelines reconstruct patient pathways from event logs.[42] These paradigms are complementary to the proposed third layer and, in many cases, provide the downstream infrastructure required to operationalize its outputs. The distinction that isolates the gap is between three kinds of process information (Table 1): *prescribed* process (what should generally happen), *observed* process (what actually happened), and *intended* process (what is supposed to happen for a specific patient). Guidelines and clinical decision support address the prescribed kind; records and process mining address the observed kind. The third layer targets the intended kind, and its defining problem is not generating clinical text but transforming what people communicate about future care into source-grounded, verifiable, executable, and monitorable clinical commitments.

Process type	Meaning	Typical source	Existing paradigm
Prescribed	What should generally happen	guideline / protocol	CIG / clinical decision support
Observed	What actually happened	event log / EHR	process mining
Intended	What is supposed to happen for this patient	clinical communication	proposed third layer

Table 1. *Three kinds of process information. The proposed third layer targets intended process, which the prescribed and observed paradigms do not recover from natural communication.*

Contributions. (1) explicit criteria for a computational layer of healthcare IT, and a three-layer model organized by the unit of computability and its enabling computational stack; (2) a formal definition of the Actionable Clinical Record (ACR), the atomic object the third layer produces, with a maturity ladder from *mentioned* to *closed*; (3) the transformation from communication to ACRs to workflow, positioning the ACR between natural clinical communication and existing structured infrastructure; (4) reliability formulated as an explicit system property, and an executable-correctness benchmark suite; (5) a feasibility instance with empirical evidence, convergent evidence from adjacent systems, testable predictions, and a four-part research agenda.

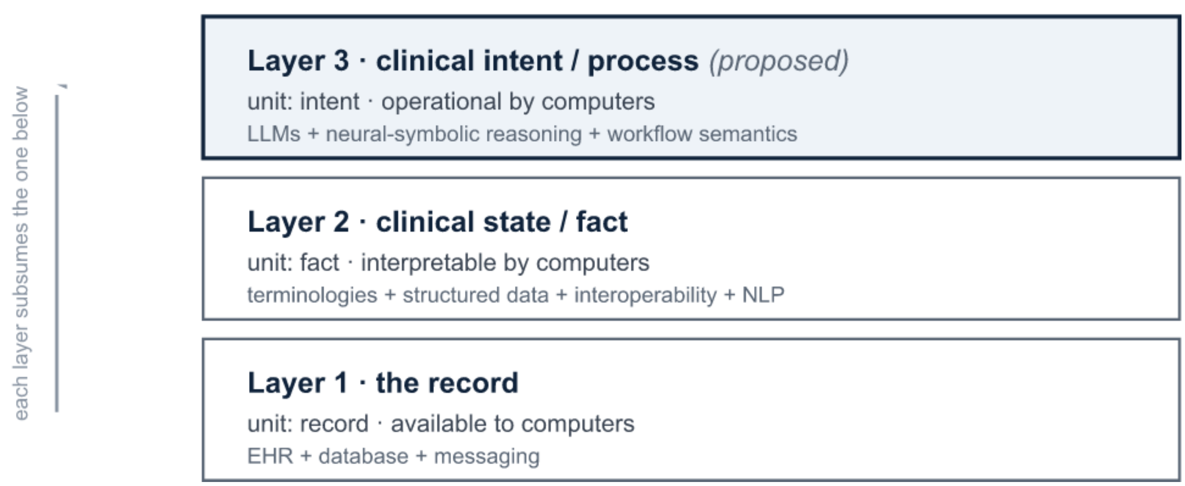

Figure 1. *The three computational layers of healthcare IT, organized by the unit of information each makes computable and its enabling computational stack. Layers accumulate rather than replace: layer 3 is proposed. Each layer answers a different question, respectively What was documented?, What is true?, and What should happen?*

2. What Defines a Computational Layer of Healthcare IT

We distinguish a generation from an incremental technological trend by requiring five structural markers: (i) a new atomic *unit of computability*; (ii) an *external driver* that makes the shift economically or clinically pressing; (iii) an *enabling technical capability*; (iv) a new *class of computation* the unit unlocks; and (v) a *residual limitation* the shift does not resolve. A proposed generation that does not satisfy these markers is better interpreted as an incremental capability within an existing layer. The framing carries one predictive claim: the residual limitation of each layer motivates the computational object addressed by the subsequent layer. Table 2 records the three layers against the markers; the markers are facets of a single shift rather than five independent tests.

Marker	Layer 1: Record	Layer 2: Facts	Layer 3: Proposed emerging layer
Unit of computability	Clinical record	Discrete clinical fact	Intended clinical action (the ACR)
External driver	HITECH / Meaningful Use policy and reimbursement	Interoperability and quality-measurement mandates	Care-continuity and patient-safety pressure
Enabling technical capability	Transactional databases and messaging	Terminologies, structured systems, interoperability, clinical NLP	Language models with symbolic reasoning and workflow semantics
Class of computation unlocked	Store, retrieve, exchange	Compute on facts (decision support, quality, research)	Compute on process (monitor, prioritize, close loops)
Residual limitation	Content is unstructured narrative	Facts are coded, but intended actions are not	Reliability, safety, evaluation, integration (open)

Table 2. *The three layers against the five markers. Layers 1 and 2 are retrospective; layer 3 is proposed. The residual limitation of each layer motivates the unit of the next.*

Related framings. Our lens is complementary to existing models. The data-information-knowledge hierarchy (DIKW) describes a generic ascent from raw data to actionable knowledge;[6] we specialize the question to *which* clinical object is made computable at each step. Maturity models such as the HIMSS Electronic Medical Record Adoption Model (EMRAM) and the wider family of hospital digital-maturity frameworks measure *how much* capability an institution has adopted;[7,8] our model is orthogonal, naming *what kind* of information becomes computable, with the third layer lying beyond the capabilities those ladders currently score. Prior work has also used generational and evolutionary language for electronic health records;[1,2] we make the organizing principle explicit and testable.

Scope of the model. The three layers describe a progression of capabilities, not a fixed global timeline: adoption differs by country and institution, and the periodization is anchored in the well-documented history of United States electronic health records. The layers are cumulative, each subsuming those beneath it. The empirical grounding offered here is a single feasibility instance (Section 8); the model's value is as an organizing lens and a source of testable predictions.

3. Layer One: Digitizing the Record and Its Transactions

The first layer digitized the clinical record and its transactions, moving from the paper chart to electronic capture, storage, retrieval, transmission, and transactional operations such as order entry and results routing. In the United States this was driven by the HITECH Act (2009) and the Meaningful Use incentive program, which moved EHR adoption from a minority of hospitals to near-universal within a decade.[9,10] The intellectual root is older: Weed's problem-oriented medical record proposed that the chart be an organized, auditable information structure rather than a chronological narrative.[11] Its enabling capability was the transactional database, without automated interpretation of clinical content: data were stored and moved, but their clinical meaning was not made computable across systems, which the second layer would address.

Layer 1: concrete examples

Systems: Epic, Oracle Health/Cerner, MEDITECH, Allscripts; the U.S. Department of Veterans Affairs VistA/CPRS; PACS for radiology images; departmental LIS and pharmacy systems.

Algorithms & tech stack: relational databases and transactional (OLTP) systems; document imaging and scanning; structured forms and templates; client-server and later web architectures; HL7 v2 messaging for point-to-point exchange; DICOM for images.

Use cases & needs: legibility and availability of the chart; e-prescribing and order entry (CPOE); charge capture and billing; regulatory reporting; basic clinical documentation. The unmet need it exposed: the record was digital but its *content* was largely unstructured narrative.

4. Layer Two: Digitizing Clinical Facts (Computable Clinical State)

The second layer made discrete clinical facts computable and exchangeable, turning the record into computable clinical *state*. Its enabling stack was broad and cannot be reduced to a single technique: structured data entry and templates, controlled terminologies and ontologies, computerized order entry and coded problem and medication lists, clinical data warehouses, and interoperability standards were at least as important as language processing.[14,15] Clinical natural-language processing was one route by which narrative facts entered this semantic layer: systems such as MedLEE and cTAKES parsed reports into coded concepts,[12,13] complementing facts captured as structured data at the point of entry. Together these enabled the first broad wave of computation *on* clinical data: decision support, quality measurement, registries, and secondary research use.[16] A persistent tension nonetheless remained: encoding clinical meaning into structured fields captures some facts, while much of the reasoning, and nearly all of the intended action, stays in narrative.[17]

Layer 2: concrete examples

Systems: structured EHR modules and clinical decision support (CDS); clinical-NLP pipelines (MedLEE, cTAKES, CLAMP); health information exchanges (HIEs); SMART on FHIR apps and terminology servers;[18] clinical data warehouses and research networks such as i2b2 and OHDSI/OMOP; eCQM quality-measure engines.

Algorithms & tech stack: rule-based and statistical clinical NLP for concept extraction; controlled terminologies and ontologies (SNOMED CT, ICD-10, LOINC, RxNorm);[14] rule-based CDS (drug-interaction checks, alerts, Arden syntax); HL7 v2/v3, CDA, and FHIR for interoperability;[15] the OMOP common data model and ETL; structured data entry; phenotyping and supervised ML on tabular EHR data.

Use cases & needs: interoperability and data exchange; decision support and safety alerting; quality and performance measurement; population health and registries; secondary use for research;[16] coded problem/medication lists and reimbursement coding. The unmet need it exposed: coding a *fact* is not the same as capturing an *action*; the care process, distributed across communication, remained outside the structured layer.

5. The Unresolved Computational Gap: Patient-Specific Intent in Clinical Communication

The first two layers made clinical *state* computable, the record and the facts it contains, but not the *intended actions* that constitute care. Follow-up instructions, conditional plans, handover items, and self-care directives are authored and consumed as language, and they fail as language: they are forgotten, mis-scheduled, or never closed.[3,17] The clinical-safety literature documents this loop-closure failure as a contributor to diagnostic error and avoidable harm, with system-level economic weight.[19,20] The gap is structural rather than incidental: the missing information is an executable representation of the intended next action, which neither the digital record nor the coded fact encodes. A single note makes the three layers concrete:

One note, three layers

Clinical note: "Repeat CBC in two weeks; refer to cardiology if chest pain persists."

Layer 1 (record): the sentence is stored verbatim as searchable text; nothing about it is computable beyond keyword search.

Layer 2 (facts): "CBC" maps to a LOINC laboratory concept and "chest pain" to a SNOMED problem, but the plan, the repeat, the referral, and their timing and condition, stays in narrative.

Layer 3 (clinical intent): two Actionable Clinical Records are emitted, each grounded in its source span and monitorable:

{action: repeat CBC, temporal: +14 days, status: pending, provenance: span#1}

{action: cardiology referral, condition: chest pain persists, actor: physician, status: conditional, provenance: span#2}

6. Layer Three: Computable Clinical Intent

The proposed third layer makes patient-specific clinical intent computable by recovering it from clinical communication and representing it as structured, executable records. Its enabling stack is modern AI: language models for semantic interpretation, coupled with symbolic reasoning for verifiable structure. The atomic object is no longer a record or a fact but the Actionable Clinical Record (ACR). We give it a formal definition, because a named, shared construct can be operationalized, evaluated, and extended by other work rather than only discussed.

6.1 The Actionable Clinical Record

Definition 1 (Actionable Clinical Record). An ACR is a source-grounded representation of a patient-specific intended clinical action together with the semantic attributes required to interpret, execute, monitor, or audit that action. Formally, it is the tuple ACR = ⟨ action, target, actor, temporal constraint, condition, dependency, status, provenance, confidence ⟩ in which *action*, *status*, and *provenance* are required (the minimum for a record that is safe to act on and to audit) and the remaining attributes are populated when the communication supports them.

To fix terminology, we use a single hierarchy throughout: clinical communication expresses clinical *intent*; an intent decomposes into atomic *actions*; each action is represented computationally as an *ACR*; and a *process* (or workflow) is a collection or sequence of ACRs and related events. The fields are defined semantically, not as free text:

ACR field semantics

action the intended clinical act (repeat a test, refer, adjust a medication, escalate). **target** the object of the action (the CBC, cardiology, the drug and new dose), linked where possible to a layer-2 coded concept.

actor the responsible party (ordering physician, specialty service, patient, nurse); may be unresolved and flagged as such. **temporal constraint** a normalized time semantics, not merely a date: absolute ("within six months"), relative ("before the next visit"), event-anchored ("after finishing antibiotics"), or conditional-recurring, any of which may resolve to a computed due date.

condition the clinical trigger or precondition ("if chest pain persists"). **dependency** ordering or prerequisite relations to other ACRs or results. **status** position on the maturity ladder defined below.

provenance the grounding of the record: source communication, source span, speaker, and encounter, so every executable item traces to the exact text that produced it. **confidence** a calibrated, per-field uncertainty used to route uncertain records to human review.

The representation extends from the minimal *action + time* pair studied in Section 8 to the full tuple (Figure 2), with provenance and confidence present on every record.

Figure 2. *The ACR schema. action, status, and provenance are required; the remaining fields are populated when the communication supports them. Every record carries a per-field confidence used for selective human review.*

The third layer is a capability ladder rather than a single capability, and separating its rungs prevents both over- and under-claiming. Recognizing that "CBC" appears in a note is layer 2. Extracting *repeat CBC* as an intended action is a partial step. Linking that action to "two weeks" and to a responsible actor produces an ACR; mapping it to a computed due date and a workflow object makes it executable; detecting completion or escalation closes the loop. We name the rungs *mentioned* → *interpreted* → *actionable* → *executable* → *monitored* → *closed*, which also draws the distinction between an *actionable* record (a well-formed intent) and an *executable* one (an intent with enough resolved information to act on automatically). Systems can be classified by the highest rung they reach reliably, and the same ladder doubles as the *status* field of the ACR.

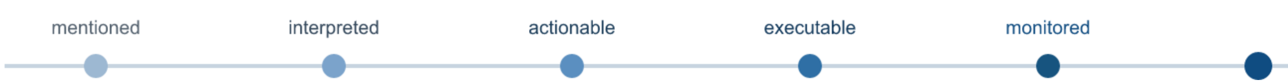

Figure 3. *The ACR maturity ladder. Recognition of concepts belongs to layer 2; the third layer begins at interpreted intent and matures toward closed-loop execution.*

6.2 Representative systems and use cases

Follow-up is the most-documented case, but it is one member of a category. Table 3 lists communication-derived process objects that the same ACR machinery targets; the systems and stack below are the settings in which these objects arise.

Process object	**Example communication**	**Salient ACR fields**
Follow-up	"Repeat CBC in two weeks."	action, temporal constraint (+14d), actor
Conditional escalation	"Return immediately if fever develops."	action, condition, actor = patient
Medication transition	"Reduce the dose after seven days."	action, target (drug/dose), temporal constraint

Process object	Example communication	Salient ACR fields
Handoff commitment	"Cardiology to review the echo tomorrow."	action, actor = service, temporal constraint
Referral	"Refer to dermatology for the lesion."	action, target, actor, dependency
Pending-result responsibility	"I will follow up on the biopsy result."	action, actor, dependency (on result)
Self-management action	"Check blood pressure daily and log it."	action, actor = patient, recurring temporal
Patient-declared intent	"I will stop taking it until I speak with my doctor."	action, actor = patient, condition

Table 3. *Communication-derived process objects targeted by the ACR. Follow-up is the empirical instance of Section 8; the others share the same schema.*

Layer 3: representative systems and stack

Systems: ambient clinical documentation / AI scribes (Nuance DAX Copilot, Abridge, Suki, Nabla) that turn the spoken encounter into notes;[21] follow-up and care-gap engines; referral-loop and results-management tools; agentic clinical-workflow assistants; the MedFollow / CareLoop line of work as a research instance.[22]

Algorithms & tech stack: clinical language models (BioBERT, ClinicalBERT, GatorTron, Med-PaLM);[23–26] clinical information extraction (MedLEE, cTAKES, CLAMP) inherited from the previous generation;[12,13,27] temporal extraction and normalization (i2b2-2012 temporal relations, THYME / Clinical TempEval, HeidelTime / SUTime);[28–32] neural-symbolic pipelines pairing learned extraction with deterministic temporal reasoning;[33] selective prediction and uncertainty calibration; event and relation extraction;[34] FHIR write-back for actionable outputs.[15]

Use cases & needs: follow-up instruction extraction; care-gap and referral-loop closure; scheduling and reminder automation; quality and safety monitoring; safe handovers; patient-message triage; turning ambient-captured encounters into executable tasks. Needs it imposes: reliability over fluency, correct temporal computation, source-linked auditability, and human oversight for uncertain cases.

6.3 From communication to workflow

The third layer functions as a bridge rather than a replacement EHR. Its input surface is the full range of clinical communication: discharge instructions, patient–clinician conversation, EMS and nurse handovers, referral correspondence, portal messages, telephone calls, and ambient encounter transcripts. Its output is a set of ACRs that maps onto infrastructure the field already has: FHIR *Task*, *ServiceRequest*, and *CarePlan*, schedulers, alerts and work queues, referral systems, and results management.[15] Existing systems can execute instructions that have already been structured; the unmet need is recovering those instructions when they originate as natural communication (Figure 4).

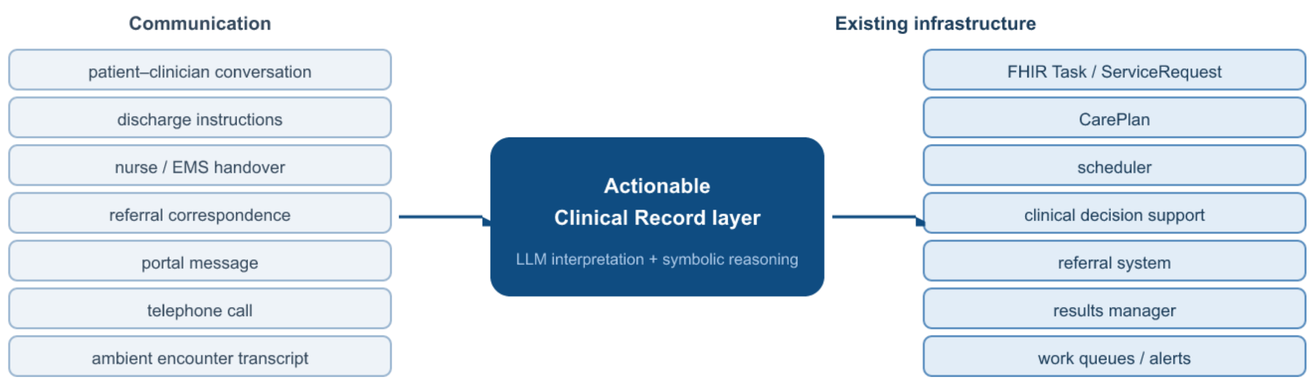

Figure 4. *The missing middle. Healthcare already has infrastructure to execute known instructions (right); the ACR layer (center) recovers patient-specific intent from natural communication (left) so it can reach that infrastructure automatically.*

6.4 Boundaries with related clinical-informatics paradigms

Two mature lines can each be mistaken for the third layer, and the distinction sharpens the claim. The third layer is not classical clinical NLP. Concept extraction from notes has existed for decades (MedLEE, cTAKES, CLAMP), but it produces *facts*: it records that "MRI" appears, feeding the layer-2 structured representation.[12,13,27] The third layer produces an executable *action*: that an MRI is due in two weeks, still pending, owned by a specific actor, at a reliability high enough to act on rather than only index. The object differs (intent, not fact), the bar differs (act, not retrieve), and recent advances in language models substantially improve the feasibility of recovering these relations from heterogeneous clinical language at scale. Improving clinical NLP is therefore not, on its own, a layer transition: the layers are cumulative, and the third is distinguished by the object it makes computable and the reliability bar it must clear, not by the technique. The third layer is also not computer-interpretable guidelines. That program encoded generic, population-level protocols as executable models;[35,36] the third layer recovers patient-specific actionable records from the actual communication of a given encounter. The two are complementary: guidelines state what should generally happen, and the third layer captures what a clinician actually instructed for this patient.

6.5 Convergent evidence

The claim does not rest on a single system; independent lines of work are converging on the actionable record. Ambient documentation has moved into routine practice, converting the spoken encounter into notes and, increasingly, into orders and tasks.[21,37] Clinical LLM evaluation is shifting toward task competence rather than fluency, with benchmarks that measure diagnostic-questioning efficiency in patient conversations and thereby treat the clinical dialogue as a measurable process.[38] Care-gap, results-management, and referral-loop tools already act on structured follow-up signals, and early agentic clinical assistants attempt multi-step workflow actions. These separate research and product lines, largely independent of one another, converge on the same object: a structured, monitorable representation of the intended next action. The feasibility instance of Section 8 sits within this convergence.

7. A Comparative View of the Three Layers

Table 4 consolidates the three layers. The upper rows state the organizing principle; the lower rows ground it in systems, stack, and value. The *Question* row is the compact form of the argument: *What was documented? → What is true? → What should happen?*

Axis	Layer 1: Record	Layer 2: Facts	Layer 3: Intent (proposed)
Object	Record	Clinical state / fact	Clinical intent / process
Question	What was documented?	What is true about the patient?	What is supposed to happen?
Representation	Documents / events	Concepts / observations	Actions, constraints, dependencies
Primary computation	Store / retrieve / exchange	Query / reason / predict	Schedule / coordinate / monitor / close
Enabling computational stack	EHR, relational databases, HL7 v2, DICOM, messaging	Terminologies, structured systems, interoperability, NLP, ML on tabular data	Language models, relation and temporal reasoning, workflow semantics, selective prediction
Representative systems	Epic, Cerner, VistA/CPRS, MEDITECH, PACS, LIS	CDS, clinical-NLP pipelines, HIE, SMART-on-FHIR apps, i2b2/OMOP warehouses, eCQM engines	Ambient scribes, follow-up and care-gap engines, agentic workflow assistants, MedFollow/CareLoop
Canonical output	Document	Fact	Actionable Clinical Record
Value unlocked	Legible, available digital substrate; billing and basic safety	Computation on data: decision support, quality measurement, research	Loop closure and follow-up safety; care coordination and workflow automation
Residual limitation	Meaning trapped in narrative	Intent trapped in communication	Safe, reliable execution

Table 4. *The three layers by object, question, representation, computation, stack, output, and residual limitation. Layer 3 is the proposed prospective layer.*

8. A Feasibility Instance: Follow-Up Extraction

Follow-up extraction is the minimal complete instance of the third layer: it takes clinical communication and emits an executable action-and-time record. A hybrid neural-symbolic pipeline, learned extraction of actions and time expressions coupled with deterministic temporal normalization, was evaluated on a controlled 2,000-note benchmark against generative baselines (Table 5).[22]

Evaluation condition	Hybrid pipeline	Generative baseline
Pair F1, seen action types	0.997	~0.53
Pair F1, unseen action types	0.986	—

Evaluation condition	Hybrid pipeline	Generative baseline
Action detection F1	>0.96	>0.96
Temporal-normalization error	0 days	—

Table 5. *Follow-up extraction on a controlled 2,000-note benchmark. Generative baselines are GPT-4o-mini and LLaMA-3.[22]*

The generative models recognized actions accurately yet performed poorly on the end-to-end task, because argument linking and temporal reasoning remained implicit in generation. The observation supporting the broader argument is that value is created not in recognizing clinical concepts but in assembling them into an executable representation, which is where explicit, verifiable structure earns its place. We state the standing of this result plainly: these results are not evidence for the three-layer model itself; they establish only the technical feasibility of one minimal ACR extraction task. The model's value rests on the ACR construct and the evaluation that measures its recovery, to which we now turn.

8.1 An executable-correctness benchmark suite

Because the third layer acts, it should be evaluated on executable correctness rather than text overlap. We propose a benchmark suite for ACR recovery, each dimension a separate, reportable measure: *action detection* (are intended actions found?); *argument extraction* (are target, actor, condition recovered?); *action–time, condition, and actor linking* (are arguments bound to the right action?); *temporal-normalization error* (days between computed and reference due dates); *executable-record exact match* (whole-ACR correctness); *unsupported-action rate* (fabricated commitments, ideally zero); *omitted-action rate* (missed commitments); *calibration and selective-risk curves* (does confidence predict correctness, and does abstention improve accuracy on the accepted set?); *source-provenance accuracy* (does each record point to the correct span?); and, ultimately, *loop-closure outcome* (does the acted-upon record reach completion?). The extension from action-and-time to the full ACR tuple is the natural next increment, motivated by real transcribed notes containing medication changes, conditional and generic follow-up, self-care, and actions without explicit timing.

9. Reliability and System Architecture

9.1 Why increasingly capable models do not remove the requirements

The third layer stands on established capabilities: clinical and general language models for semantic interpretation of heterogeneous clinical language;[23–26] clinical information extraction for entities and relations (MedLEE, cTAKES, CLAMP, and the i2b2/n2c2 benchmarks);[12,13,27,34] temporal reasoning for the calendar arithmetic that turns "in two weeks" into a date;[28–32] and ambient documentation, which extends the input surface from written notes to the spoken encounter.[21,37] A more capable generator does not, however, remove the requirements the layer imposes. Auditability, verifiable temporal correctness, and a calibrated signal of when to defer are properties of the *output contract* rather than gaps in model capability: a system that acts on patient trajectories must establish that a date is correct and trace it to its source, however fluent the underlying model becomes. General-purpose models interpret clinical language well but remain unreliable precisely where the layer is most demanding, in linking an action to its time and normalizing it into an executable schedule without omission or fabrication.[39,40]

9.2 Neural-symbolic decomposition

These requirements motivate a decomposition in which learned components handle interpretation and explicit components handle verifiable structure.[33] The argument for explicit reasoning is specific rather than ideological: some workflow semantics, calendar arithmetic, dependency ordering, and status transitions, are formally specified, and explicit symbolic components can provide deterministic or independently verifiable transformations where those semantics hold. Such semantics need not remain latent inside generative inference. Learned extraction supplies the candidate actions and arguments; deterministic reasoning resolves times, orders dependencies, and computes due dates; and selective prediction routes uncertain cases to a human. Reliability, not scale, is the binding constraint on this architecture.

9.3 Provenance, calibration, and oversight

We formulate reliability as an explicit system property: a set of invariants a care-process system should satisfy, each checkable and each shaping the design. *(I1) No executable item without provenance:* every acted-upon ACR traces to a source span, speaker, and encounter. *(I2) Deterministic computation where deterministic semantics exist:* calendar arithmetic and dependency ordering are computed, not generated. *(I3) Uncertainty on every inferred field:* a calibrated confidence accompanies each value. *(I4) Abstention:* the system may decline rather than guess. *(I5) Risk-proportionate human review:* higher-consequence actions receive tighter oversight. *(I6) An immutable trace from communication to action,* so any executed commitment can be reconstructed and audited.[41] *(I7) An explicit boundary between extracted and inferred content:* what the clinician stated versus what the system concluded. These invariants connect the third layer to the older program of computer-interpretable guidelines and care pathways, which sought executable models of process but lacked a reliable path from free-text communication into those models.[35,36]

10. Predictions and the Computable Care Process Research Agenda

10.1 Testable predictions

If the three-layer model is correct, the following are checkable over the next several years, and their failure would count against it:

P1. The machine-actionable use of existing FHIR workflow resources (*Task*, *CarePlan*, *ServiceRequest*) will grow: today used mainly for documentation and exchange, they will increasingly carry executable, closed-loop follow-up, and the standard will extend their action and temporal semantics accordingly. *Checkable by* a rising share of follow-up actions represented as these resources rather than free text (horizon 3–5 years).

P2. Evaluation will shift from text-overlap metrics to *executable correctness*: action–time pair linking, temporal accuracy, and loop-closure rate. *Checkable by* new clinical benchmarks reporting these measures rather than ROUGE/BLEU-style overlap (horizon 2–4 years).

P3. Reliable systems will converge on *source-linked, auditable* action records with selective human review, rather than end-to-end free generation. *Checkable by* deployed follow-up and results-management tools exposing the source span and a confidence-based review queue (horizon 3–5 years).

P4. Ambient documentation will move downstream from producing notes to producing *tasks and orders*, closing the loop into scheduling and results management. *Checkable by* ambient-scribe products adding order and task generation and results follow-up, not just note drafting (horizon 2–4 years).

10.2 The Computable Care Process research agenda

Beyond the predictions, the framework defines a menu of publishable problems in four research domains, so that a system or study can locate itself precisely.

D1. Representation. A consensus ACR ontology: an extensible taxonomy of clinical actions, required-versus-optional field semantics, temporal and conditional constraint types, provenance, and reconciliation when messages conflict, revise, or revoke a plan.

D2. Inference and reliability. ACR recovery from multilingual and multimodal communication; relation construction and temporal reasoning over event-anchored and conditional constraints; cross-message reconciliation; calibrated uncertainty and abstention; contradiction handling; and distinguishing patient intent from clinician intent.

D3. Evaluation. Authentic-data benchmarks beyond synthetic corpora; the executable-correctness suite of Section 8.1 adopted across tasks; cross-institution and cross-language generalization; and measures of clinical utility.

D4. Integration and impact. ACR-to-FHIR mappings and write-back into scheduling, results-management, and referral systems;[15,35] human-factors design of the review queue; regulatory classification of care-process software; and whether automatic loop closure measurably improves clinical outcomes.

Moving from synthetic corpora to de-identified real notes across languages and documentation styles remains the central empirical risk, and safe deployment presupposes the provenance, abstention, and oversight invariants of Section 9.

11. Conclusion

Read through the unit of information it makes computable, healthcare IT progresses from the record to the clinical fact and, we propose, to patient-specific clinical intent. Each layer is defined by that unit, record, then fact, then intent, and by the computational stack that supports it, and the layers accumulate rather than replace one another. The problem that characterizes the third layer is not generating clinical text but transforming what people communicate about future care into source-grounded, verifiable, executable, and monitorable clinical commitments, represented as the Actionable Clinical Record. The layer does not replace existing workflow infrastructure; it recovers the intent that lets natural clinical communication reach that infrastructure, under reliability formulated as an explicit system property. A feasibility instance shows that the minimal case is within reach, and convergent developments suggest the layer is emerging from several directions at once. The ACR definition, maturity ladder, benchmark suite, predictions, and research agenda are offered as constructs that can be operationalized, evaluated, extended, or falsified in subsequent work.

References

1. Ammenwerth E, et al. Twenty-Five Years of Evolution and Hurdles in Electronic Health Records and Interoperability in Medical Research: Comprehensive Review. *J Med Internet Res* 2025;27:e59024. doi:10.2196/59024.

2. Evans RS. Electronic Health Records: Then, Now, and in the Future. *Yearb Med Inform* 2016;25(Suppl 1):S48–S61. doi:10.15265/IYS-2016-s006.

3. Callen JL, Westbrook JI, Georgiou A, Li J. Failure to follow-up test results for ambulatory patients: a systematic review. *J Gen Intern Med* 2012;27(10):1334–1348. doi:10.1007/s11606-011-1949-5.

4. American College of Radiology. Recommendations Follow-Up Improvement Collaborative (accessed 2026).

5. Closing the referral loop: primary-care referrals to specialists in a large health system. Duke Institute for Health Innovation; CMS eCQM CMS50v13.

6. Rowley J. The wisdom hierarchy: representations of the DIKW hierarchy. *J Inf Sci* 2007;33(2):163–180. doi:10.1177/0165551506070706.

7. Li R, Niu Y, Scott SR, et al. Using Electronic Medical Record Data for Research in a HIMSS Analytics EMRAM Stage 7 Hospital in Beijing: Cross-sectional Study. *JMIR Med Inform* 2021;9(8):e24405. doi:10.2196/24405.

8. Duncan R, Eden R, Woods L, Wong I, Sullivan C. Synthesizing Dimensions of Digital Maturity in Hospitals: Systematic Review. *J Med Internet Res* 2022;24(3):e32994. doi:10.2196/32994.

9. Blumenthal D, Tavenner M. The “Meaningful Use” Regulation for Electronic Health Records. *N Engl J Med* 2010;363:501–504. doi:10.1056/NEJMp1006114.

10. Adler-Milstein J, et al. Electronic Health Record Adoption In US Hospitals: Progress Continues, But Challenges Persist. *Health Aff* 2015;34(12):2174–2180. doi:10.1377/hlthaff.2015.0992.

11. Weed LL. Medical Records That Guide and Teach. *N Engl J Med* 1968;278:593–600, 652–657. doi:10.1056/NEJM196803212781204.

12. Friedman C, Alderson PO, Austin JHM, Cimino JJ, Johnson SB. A general natural-language text processor for clinical radiology (MedLEE). *J Am Med Inform Assoc* 1994;1(2):161–174. doi:10.1136/jamia.1994.95236146.

13. Savova GK, et al. Mayo clinical Text Analysis and Knowledge Extraction System (cTAKES). *J Am Med Inform Assoc* 2010;17(5):507–513. doi:10.1136/jamia.2009.001560.

14. Bodenreider O, Cornet R, Vreeman DJ. Recent Developments in Clinical Terminologies — SNOMED CT, LOINC, and RxNorm. *Yearb Med Inform* 2018;27(1):129–139. doi:10.1055/s-0038-1667077.

15. Bender D, Sartipi K. HL7 FHIR: An Agile and RESTful Approach to Healthcare Information Exchange. *IEEE CBMS* 2013:326–331. doi:10.1109/CBMS.2013.6627810.

16. Safran C, Bloomrosen M, Hammond WE, et al. Toward a National Framework for the Secondary Use of Health Data: An AMIA White Paper. *J Am Med Inform Assoc* 2007;14(1):1–9. doi:10.1197/jamia.M2273.

17. Rosenbloom ST, Denny JC, Xu H, et al. Data from clinical notes: a perspective on the tension between structure and flexible documentation. *J Am Med Inform Assoc* 2011;18(2):181–186. doi:10.1136/jamia.2010.007237.

18. Mandel JC, Kreda DA, Mandl KD, Kohane IS, Ramoni RB. SMART on FHIR: a standards-based, interoperable apps platform for electronic health records. *J Am Med Inform Assoc* 2016;23(5):899–908. doi:10.1093/jamia/ocv189.

19. Singh H, Meyer AND, Thomas EJ. The frequency of diagnostic errors in outpatient care. *BMJ Qual Saf* 2014;23(9):727–731. doi:10.1136/bmjqs-2013-002627.

20. Slawomirski L, Kelly D, de Bienassis K, Kallas K-A, Klazinga N. The economics of diagnostic safety. *OECD Health Working Papers* 2025. doi:10.1787/fc61057a-en.

21. Ambient AI Scribes in Clinical Practice: A Randomized Trial. *NEJM AI* 2025. doi:10.1056/AIoa2501000.

22. Laufer M, Aperstein Y, Apartsin A. Reliable Extraction of Clinical Follow-Up Instructions: A Hybrid Neural-Symbolic Pipeline. *arXiv* 2026:2605.26560.

23. Lee J, et al. BioBERT: a pre-trained biomedical language representation model. *Bioinformatics* 2020;36(4):1234–1240. doi:10.1093/bioinformatics/btz682.

24. Alsentzer E, et al. Publicly Available Clinical BERT Embeddings. *Proc. 2nd Clinical NLP Workshop* 2019:72–78. arXiv:1904.03323.

25. Yang X, et al. A large language model for electronic health records (GatorTron). *npj Digit Med* 2022;5:194. doi:10.1038/s41746-022-00742-2.

26. Singhal K, et al. Large language models encode clinical knowledge (Med-PaLM). *Nature* 2023;620:172–180. doi:10.1038/s41586-023-06291-2.

27. Soysal E, et al. CLAMP: a toolkit for efficiently building customized clinical NLP pipelines. *J Am Med Inform Assoc* 2018;25(3):331–336. doi:10.1093/jamia/ocx132.

28. Sun W, Rumshisky A, Uzuner O. Evaluating temporal relations in clinical text: 2012 i2b2 Challenge. *J Am Med Inform Assoc* 2013;20(5):806–813. doi:10.1136/amiajnl-2013-001628.

29. Styler WF, et al. Temporal Annotation in the Clinical Domain (THYME). *Trans Assoc Comput Linguist* 2014;2:143–154. doi:10.1162/tacl_a_00172.

30. Bethard S, et al. SemEval-2016 Task 12: Clinical TempEval. *Proc. SemEval-2016* 2016:1052–1062. doi:10.18653/v1/S16-1165.

31. Strötgen J, Gertz M. Multilingual and cross-domain temporal tagging (HeidelTime). *Lang Resour Eval* 2013;47(2):269–298. doi:10.1007/s10579-012-9179-y.

32. Chang AX, Manning CD. SUTime: A library for recognizing and normalizing time expressions. *Proc. LREC* 2012:3735–3740.

33. Garcez A, Lamb LC. Neurosymbolic AI: the 3rd wave. *Artif Intell Rev* 2023;56:12387–12406. doi:10.1007/s10462-023-10448-w.

34. Uzuner O, South BR, Shen S, DuVall SL. 2010 i2b2/VA challenge on concepts, assertions, and relations in clinical text. *J Am Med Inform Assoc* 2011;18(5):552–556. doi:10.1136/amiajnl-2011-000203.

35. Peleg M. Computer-interpretable clinical guidelines: a methodological review. *J Biomed Inform* 2013;46(4):744–763. doi:10.1016/j.jbi.2013.06.009.

36. Lenz R, Reichert M. Computerization of workflows, guidelines, and care pathways. *J Am Med Inform Assoc* 2011;18(6):738–748.

37. Clinician perspectives on ambient AI scribes. *J Am Med Inform Assoc* 2026;33(2):255.

38. Werthaim M, Kimhi M, Apartsin A, Aperstein Y. A benchmark for evaluating diagnostic questioning efficiency of LLMs in patient conversations. *Sci Rep* 2026;16:6121. doi:10.1038/s41598-026-37022-y.

39. Ji Z, et al. Survey of Hallucination in Natural Language Generation. *ACM Comput Surv* 2023;55(12):248. doi:10.1145/3571730.

40. Nori H, et al. Capabilities of GPT-4 on Medical Challenge Problems. *arXiv* 2023:2303.13375.

41. Aperstein Y, Gottlib A, Benita G, Apartsin A. Explainable Semantic Text Relations: A Question-Answering Framework for Comparing Document Content. *Information* 2025;16(12):1090. doi:10.3390/info16121090.

42. van der Aalst W. *Process Mining: Data Science in Action*. 2nd ed. Berlin: Springer; 2016. doi:10.1007/978-3-662-49851-4.